



Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058-India
(Autonomous College Affiliated to University of Mumbai)

Mid Semester Examination
March 16, 2018

Max. Marks:- 30
Class:- T.E.
Course Code:- ETC605
Name of the Course:- Operating Systems

Duration:- 1Hr 30 Min.
Semester:- VI
Branch:- Electronics & Telecommunication

Q.1:- The following table shows waiting time and turnaround time for FCFS, SJF, non preemptive priority and RR (quantum time=1) scheduling policies for the given process sequence.

PROCESS	CPU BURST TIME	PRIORITY	FCFS		SJF		PRIORITY		RR (QUANTUM=1)	
			TW	TA	TW	TA	TW	TA	TW	TA
P1	10	3	0	10	9	19	6	16	9	19
P2	1	1	10	11	0	1	0	1	1	2
P3	2	3	11	13	2	4	16	18	5	7
P4	1	4	13	14	1	2	18	19	3	4
P5	5	2	14	19	4	9	1	6	9	14
AVERAGE			9.6		3.2		8.2		5.4	

10 marks for four scheduling algorithms.

Q.2:- Process states:- New, Ready, Run, Blocked, Halted

Transitional activities to be listed for state transition with the help of diagram....5 Marks

Q.3:- USER: User interface, Program execution, I/O operation, File system manipulation, Communications, Error detection

SYSTEM: Resource allocation, accounting, Protection and security....5 Marks

Q.4:- All the requirements for 5 Marks

Q.5:- All possible differential points for 5 Marks